- Within the period from 30 days before the event to 15 days before the event, receive a notification every 5 days.
- Within the period from 15 days before the event till the event time, receive a notification every 3 days.

Note that for the CPI certificate expiry scenario, the notifications stop as soon as the certificate that will expire is deleted in the System Connections app.

In the same example, if you deactivate the "30 days" period, as shown below:

Notification Period	Frequency	Activated for User
60 days	Every 10 days	Yes
30 days	Every 5 days	**No**
15 days	Every 3 days	Yes

Then the "30 days" period is ignored and the schedule changes to:

- Within the period from 60 days before the event to 15 days before the event, receive a notification every 10 days.
- Within the period from 15 days before the event till the event time, receive a notification every 3 days.

Post-Event Notifications

Post-event notifications refer to the notifications you receive after the target event happens, for example, after your CPI certificate has expired. You can schedule post-event notifications by adding notification periods and specifying notification frequency within each period, as described below:

- Notification period: The period within which email notifications will be sent in a specified frequency.

  You only specify the end of the period, for example, **30 days after the event**, which means the notification period starts from the end of the previous period or the event time, to 30 days after the event.

- Notification frequency: The frequency to receive notifications within the specified period. For example, setting the frequency to **every 5 days** means within the specified period, you'll receive an email notification every 5 days.

Example

The schedule shown below

Notification Period	Frequency	Activated for User
15 days	Every 3 days	Yes
30 days	Every 5 days	Yes
60 days	Every 10 days	Yes

specifies the following schedule:

- Within the period from the event time to 15 days after the event, receive a notification every 3 days.
- Within the period from 15 days after the event to 30 days after the event, receive a notification every 5 days.
- Within the period from 30 days after the event to 60 days after the event, receive a notification every 10 days.

Note that for the CPI certificate expiry scenario, the notifications stop as soon as the expired certificate is deleted in the System Connections app.

In the same example, if you deactivate the "30 days" period, as shown below:

Notification Period	Frequency	Activated for User
15 days	Every 3 days	Yes
30 days	Every 5 days	**No**
60 days	Every 10 days	Yes

Then the "30 days" period is ignored and the schedule changes to:

- Within the period from the event time to 15 days after the event, receive a notification every 3 days.
- Within the period from 15 days after the event to 60 days after the event, receive a notification every 10 days.

Procedure

1. Launch the **Manage Notification Settings** app.
2. In the scenario list, select the scenario for which you want to set notifications.
3. To activate on-event notification, for **On-Event Notification**, set the toggle to **Yes**. To deactivate, set it to **No**.
4. To configure pre-event or post-event notifications, do the following:
    a. Expand the **Pre-Event Notifications** or **Post-Event Notifications** area.
    b. To add a notification period, choose the **Add** button. Then in the **Add Notification Period** dialog box, specify the start (pre-event) or end (post-event) of the period and the frequency, and choose **Save**. You can add multiple periods according to your needs.
    c. To edit or delete a notification period, choose the **Edit** or **Delete** button. To deactivate a period, set **Activated for User** to **No**.

Assign Codes

Note

To use this app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section of the Administration Guide for Shippers.

SAP Business Network for Logistics offers out-of-the-box connectivity with SAP Transportation Management (SAP TM) standalone or Transportation Management (TM) embedded in SAP S/4HANA. As a shipper, you can maintain mappings for LBN-defined code lists (such as packaging material type). When a business document is sent from a TM system to SAP Business Network for Logistics, the LBN system converts the code lists to LBN standard and displays the correct value on the UI of the corresponding app.

Note

SAP Business Network for Logistics only supports a one to one mapping of codes.

You have the following possibilities to define your code list mappings:

- Take over standard LBN code lists and their values
- Upload a spreadsheet with your customized code list values
- Add values manually

ℹ Note

You can also combine the three possibilities.

SAP Business Network for Logistics provides the following standard code lists:

- Packaging material type
  - Used only in B2B communication and not shown on the user interface (UI)
- Document reference type codes
  - Used in the **Confirm Freight Bookings** app under **Reference Documents**
  - Used in the **Freight Orders for Confirmation** app under **Reference Documents**
  - Used in the **Track Freight Movement** app under **Reference Documents**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Document Types for Business Transaction**
- Shipping type
  - Used in the **Confirm Freight Bookings** app under **Shipping Type**
  - Used in the **Track Freight Movement** app under **Shipping Type**
  - Used in freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Subcontracting** > **Shipping Type**

  The default shipping type is initially empty. You can edit the code mapping and assign a default shipping type. When a business document without a shipping type reaches SAP Business Network for Logistics, it will then assign the default shipping type. For instance, if the default shipping type is full truck load (FTL), the document's shipping type will be designated FTL.
- Transport means description

  For information about the existing values, see: Transport Means Standard Code Descriptions.
  - Used in the **Freight Orders for Confirmation** app under **Information** > **Active Vehicle Resource** > **Transport Means Description**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Means of Transport**
- Transport means type

  For information about the existing values, see: Means of Transport.
  - Used in the **Confirm Freight Bookings** app under **Information** > **Active Vehicle Resource** > **Transport Means Type**
  - Used in the **Freight Orders for Confirmation** app under **Information** > **Active Vehicle Resource** > **Transport Means Type**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Means of Transport**
- Document reference item type codes
  - Used in API communication to carrier
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Business Transaction Item Type Code**
- Contract condition code

  - Used in the **Confirm Freight Bookings** app under **Contract Condition Code**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Movement Type Code**

- Unit of measure for length
  - Used in the **Confirm Freight Bookings** app under **Cargo Item Length, Height and Width**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Item Length, Height and Width**

- Unit of measure for quantity
  - Used in the **Confirm Freight Bookings** app under **Cargo Item Quantity**
  - Used in the **Freight Orders for Confirmation** app under **Cargo Item Quantity**
  - Used in the **Track Freight Movement** App under **Cargo Item Quantity**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Item Quantity**

- Unit of measure for volume
  - Used in the **Confirm Freight Bookings** app under **Gross Volume**
  - Used in the **Freight Orders for Confirmation** app under **Gross Volume**
  - Used in the **Track Freight Movement** app under **Cargo Volume**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Gross Volume**

- Unit of measure for weight
  - Used in the **Confirm Freight Bookings** app under **Gross Weight**
  - Used in the **Freight Orders for Confirmation** app under **Gross Weight**
  - Used in the **Track Freight Movement** app under **Cargo Weight**
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Gross Weight**

- Note type
  - Used in API communication to carrier
  - Used in the freight orders and freight bookings in SAP Transportation Management standalone or Transportation Management embedded in SAP S/4HANA under **Text Type**

- Transportation mode
  - Used in the **Confirm Freight Bookings** app under **General** > **Mode of Transport**

    Used to determine the outbound API relevance from SAP Business Network for Logistics based on the entries in the **Configure Partner Connections** app.

- Carrier LBN ID

  You can assign the system ID of an external system and the internal ID of the carrier from that external system to a carrier LBN ID. To do so, choose **Add**, enter the system ID and the internal ID in the free text fields, and select a carrier LBN ID from the input help.

Downloading and Uploading Code Lists

Overview

You can download a spreadsheet with the standard code lists from SAP Business Network for Logistics, enter the required code list values to be mapped and then upload your customized spreadsheet to LBN. SAP Business Network for Logistics provides a standard spreadsheet for each code list.

Prerequisites

Only spreadsheets with the *.xlsx format are supported.

Procedure

To upload your code lists to SAP Business Network for Logistics, proceed as follows:

1. In the **Assign Codes** app, on the left-hand side, choose the code list whose mappings you want to edit.
2. Mark all standard LBN codes and delete them. (Optional)
3. Download the spreadsheet with the standard LBN codes.
4. Edit the spreadsheet according to your needs. There are two sheets with columns described below.

   Sheet 1 contains the mapping of source code to LBN code.

   - **Source Code**: The source code from the external system, such as a TM system.
   - **LBN Code**: The LBN code (such as CIF – Cost and Freight).
   - **Deletion Indicator**: Provide value Y if this mapping should be deleted and N to include or retain it.

   Sheet 2 contains the code list with LBN codes and corresponding descriptions.

   > **Note**
   > Ensure that no personal data enters the system in an uncontrolled or non-purpose-related way in the Microsoft Excel file.

5. Upload your spreadsheet to LBN.
6. Upon saving, LBN checks your entries and displays any errors or mismatches found during validation.
7. Solve the errors.
8. If required, you can add values manually.
9. Save your entries.

Assign Charge Codes

You use this app to perform the following:

- In a carrier invoicing scenario, you can assign charge types and condition types that you use in your transportation system to a set of predefined charge codes in the freight collaboration option for SAP Business Network for Logistics.

  Alternatively, you can also create and use a custom charge code.

- In a self-billing scenario, you can assign charge types that you use in your transportation system to a set of predefined charge codes in the freight collaboration option.

Alternatively, you can also create and use a custom charge code.

Because the charge types in freight orders play an important role in the calculation of transportation charges, it is important that these charge types are easily identifiable. When you use unified charge codes in the freight collaboration option, the carriers can easily interpret the charge types specific to your transportation system in freight orders.

ℹ Note

To use this app, you must add one or several roles to your role collection. The required role templates and application identifiers are described in the Available Roles for Shippers section of the Administration Guide for Shippers.

Prerequisites

If you are using Transportation Management (TM) on SAP S/4 HANA 1809 onwards, you have made the following basic settings in your TM system.

- You must use the `InvoiceRequest_In` interface to process a carrier invoice that refers to a document such as a freight order, a service order, or a freight booking in TM. When the invoice is submitted and sent from the freight collaboration option to TM, the charge codes are converted to condition types to suit the message interface `InvoiceRequest_In`.
- To enable the carrier to submit an invoice, you must maintain the following Customizing activities:
  - In Customizing for **Transportation Management** under **Integration** > **Enterprise Services** > **Carrier Invoice Integration** > **Assign Carrier Condition type to SAP TM Charge Type**
  - In Customizing for **Transportation Management** under **Integration** > **Enterprise Services** > **Carrier Invoice Integration** > **Assign Carrier Tax Types to SAP TM Tax Charge Type**

  Note that if you do not maintain these Customizing activities, the system does not create an invoice in the TM system.

You can use the **Assign Charge Codes** app to map the following entities based on the scenario type:

- Charge types

  These are the charge types that you have agreed with your carriers in the freight agreement. You maintain these charge types in your transportation system.

- Condition types

  These are the condition types which the carrier sends in the invoice via business to business (B2B) communication method. You map these condition types in your transportation system with the charge types.

- Charge codes

  You can choose a charge code that is predefined by the freight collaboration option charge codes from the input help.

  Alternatively, you can choose a custom code that you have created in the **Assign Charge Codes** app.

  You can also create a custom code from the input help. To do so, perform the following:

  1. Choose **Create Custom Code** from the input help.

     A create custom code dialog box appears.

  2. Enter a unique custom code that begins with Y or Z. Note that it must not contain any spaces or special characters, except underscore.

  3. Enter the language in which you want to provide a description.

     Note that you can add multiple descriptions in different languages after you create a custom code, using the Manage Custom Codes feature.

4. Choose **Save and Select**.

   A new custom code has been created and it has been assigned in the charge type mapping.

> **Note**
> You cannot assign multiple combinations of charge types and condition types to a single charge code. If you cannot find a predefined code, you can create a new custom code.

- Enabled for carrier

  If you select this checkbox, the system displays the charge type to the carrier while creating an unplanned charge in an invoice. If you deselect this checkbox, the system does not display the charge type to the carrier.

Carrier Invoicing Scenario

The carrier generates the invoice in their transportation system and submits the invoice via the freight collaboration option. The charges in their transportation system invoice are represented by condition types. These condition types are mapped to specific charge types in your transportation system.

When you map the charge types and condition types to charge codes, in the freight collaboration option, the carrier views the charge codes in the charges table in the freight order and the invoice documents. These charge codes provide a uniform representation of the charge types across the entire network. Hence, you must map your charge types and condition types to charge codes in the **Assign Charge Codes** app.

> **Note**
> If you are using TM on SAP S/4 HANA 1809 onwards, the charge type and condition type values that you maintain in the Customizing activity **Assign Carrier Condition type to SAP TM Charge Type** must match the values that you specify in the **Assign Charge Codes** app. If the values do not match, the inbound and outbound communication between the invoice and the freight order will be erroneous.

However, when the carrier submits and sends the invoice to your transportation system, you must receive condition types in place of business network charge codes. To enable this, replaces the business network charge codes with the corresponding charge types and condition types specific to your transportation system.

> **Note**
> If you are using TM on SAP S/4 HANA 1809 onwards and a freight order is sent from TM system to the freight collaboration option, the TM charge types are converted to charge codes. In the carrier tenant, the system fetches these charge codes from your tenant and the charge types and condition types are converted to the respective charge codes. When the invoice is submitted and sent from the freight collaboration option to TM, the charge codes are converted to condition types to suit the message interface `InvoiceRequest_In`.
>
> For more information on processing a carrier invoice using a B2B interface, see SAP Library for **SAP S/4HANA** on SAP Help Portal at https://help.sap.com/s4hana. In SAP Library, under **Product Assistance**, select the preferred language and choose **Enterprise Business Applications** > **Supply Chain** > **Transportation Management (TM)** > **Settlement** > **Freight Settlement** > **Dispute Management for Freight Settlements** > **Carrier Invoice** > **Processing of a Carrier Invoice Using a B2B Interface**.

Self-Billing Scenario

In a self-billing scenario, you must map the charge types that the shipper maintains in the TM system to standard freight collaboration option charge codes. You could download the charge types from the charge type customizing from TM and upload the same in the freight collaboration option in this app. Please note that the first row is ignored by the system during upload. Similarly, only the first column is read by the app.

Note

- Only after you assign charge codes to charge types and condition types that you use in your transportation system, can you create freight orders and invoices.
- When you choose the **Scenario** as **Both** in the **Assign Charge Codes** app, you can add a mapping common to both the carrier invoicing and self-billing scenarios.
- As of now, the freight collaboration option only supports these assignments only for the road mode of transportation.

Manage Codes

Custom Codes

When you choose the **Manage Codes** > **Custom Codes** button in the **Assign Charge Codes** app, you can view, edit, and create custom codes.

Creating a Custom Code

To create a custom code, perform the following:

1. Choose the **Create** button on the top right corner of the screen.

   A create custom code dialog box appears.

2. Enter a unique custom code that begins with Y or Z. Note that it must not contain any spaces or special characters, except underscore.

3. Enter the language in which you want to provide a description.

   Note that you can create multiple descriptions in different languages after you create a custom code.

4. Choose **Create**.

   A new custom code has been created

Adding, Editing, and Deleting Descriptions

To add a new description in a different language, choose the add icon in the **Actions** column. You can also delete or edit an existing description using the edit and delete icons respectively in the **Actions** column. You can revert the deletion of a description using the undo icon in the **Actions** column.

Note that you must maintain at least one description for a custom code in any language.

Deleting a Custom Code

To delete a custom code, you can use the delete icon in the **Actions** column.

Note

- You can only delete a custom code that does not have the **In Use** indication. This indication appears when you have used the custom code in a charge type mapping.
- If you want to delete a custom code with **In Use** indication, you must first delete the charge type mapping in the worklist page and then delete the custom code.

Saving the Changes

When you create or delete a custom code, or when you add, modify, or delete a description, you must choose Save on top right corner of the screen to save your changes.

Tax Codes

When you choose the Manage Codes > Tax Codes button in the Assign Charge Codes app, you can view, edit, and create tax codes.

The carrier can you use these tax codes and percentage values in their invoicing app when they add a tax line.

> **Note**
>
> In the General Settings app, a new toggle Enable Tax Code Validation has been introduced. When you enable this setting, if the carrier enters a tax code in the invoice, the system validates this tax code with the tax codes that you have maintained here in the Assign Charge Codes app.

Adding a Tax Code

To create a tax code, perform the following:

1. Choose the Add button on the top right corner of the screen.

   A new tax line appears.

2. Enter a tax code of up to 3 characters.

3. Enter a tax percentage and choose Save.

   A new tax code has been created.

Deleting a Tax Code

To delete a tax code, you can use the delete icon. Note that once you delete a tax code, you can undo the delete action by using the undo icon.

Saving the Changes

When you create or delete a tax code, you must choose Save on top right corner of the screen to save your changes.

Uploading and Downloading Charge Code Assignments

Downloading Charge Type Assignments

You can download the charge types and condition types assignments in the Assign Charge Codes app in a Microsoft Excel file using the Download button.

The downloaded Microsoft Excel file contains the charge code assignments in the first sheet. You can view the charge codes description in the second sheet.

Uploading Charge Type Assignments

If you have already maintained the charge types and condition types mapping in your transportation system, you can avoid the effort of manually entering them in the Assign Charge Codes app. You can upload the charge types and condition types mapping from your transportation system in the Assign Charge Codes app using the Upload button.

ℹ Note

You can only upload Microsoft Excel files of xlsx format. The Microsoft Excel file that you want to upload must be of the same format as the file that you download from the **Assign Charge Codes** app. To avoid any formatting changes, you can choose to download, edit and then upload the Microsoft Excel file here.

You can do the following:

1. **Scenario Code**

   Enter scenario code referring to the possible values provided in the column header.

2. **Charge Code (SAP Business Network for Logistics)**

   Enter the appropriate charge code by referring to the second sheet.

3. **Enabled for Carrier**

   If you select this checkbox, the system displays the charge type to the carrier while creating an unplanned charge in an invoice. If you deselect this checkbox, the system does not display the charge type to the carrier.

Downloading Charge Type Assignments from TM on SAP S/4 HANA System

If you are using TM on SAP S/4 HANA 1809 onwards, you can download the charge type and condition types mapping in Microsoft Excel format using the following procedure:

- In Customizing for **Transportation Management** under **Integration** > **Enterprise Services** > **Carrier Invoice Integration** > **Assign Carrier Condition type to SAP TM Charge Type**, choose **Table View** > **Print**.
- In the print view, choose the spreadsheet icon.

  A select spreadsheet dialog box appears.
- Choose **Select from All Available Formats** > **Excel – Office Open XML Format (XLSX)** and choose **Continue**.
- Save the file.

File Format and Contents

- The system allows you to upload only Microsoft Excel files of xlsx format.
- The first column in the Microsoft Excel file must contain the condition types and the second column must contain the charge types.
- The system ignores the first row of the Microsoft Excel file.

ℹ Note

- You must make sure that no personal data enters the system in an uncontrolled or non purpose-related way in the Microsoft Excel file.
- If the uploaded Microsoft Excel file contains mappings from your transportation system that exists already in the **Assign Charge Codes** app, the system ignores such entries and uploads only new entries from the file. The system adds the new entries to the app along with the existing entries.
- The **Assign Charge Codes** app contains a charge type to charge code mapping for a self-billing scenario and the uploaded Excel file contains a mapping for the same charge type with condition type for the carrier invoicing scenario. In this case, the system changes the scenario as **Both** in the app and includes the condition type mapping as well to this entry.
- If the uploaded Microsoft Excel file contains a different mapping for a charge type or a condition type that exists already in the app, the system does not upload the entire file.